Stat 6390.001: High Dimensional Probability and Statistics, Fall 2026 **Topic 22**

Matrix analysis and tail conditions for random matrices

Lecturer: Yun Wei *Scribe: Chin-Ling Chang, Yizhe He*

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1 Background on matrix analysis

We begin by introducing some additional background on matrix-valued functions. Recall the class $\mathcal{S}^{d\times d}$ of of symmetric $d\times d$ matrices. Any function $f:\mathbb{R}\to\mathbb{R}$ can be extended to a map from the set $\mathcal{S}^{d\times d}$ to itself in the following way. Given a matrix $\mathbf{Q}\in\mathcal{S}^{d\times d}$, consider its eigendecomposition $\mathbf{Q}=\mathbf{U}^{\mathrm{T}}\mathbf{\Gamma}\mathbf{U}$. Here the matrix $\mathbf{U}\in\mathbb{R}^{d\times d}$ is a unitary matrix, satisfying the relation $\mathbf{U}^{\mathrm{T}}\mathbf{U}=\mathbf{I}_d$, whereas $\mathbf{\Gamma}:=\mathrm{diag}(\gamma(\mathbf{Q}))$ is a diagonal matrix specified by the vector of eigenvalues $\gamma(\mathbf{Q})\in\mathbb{R}^d$. Using this notation, we consider the mapping from $\mathcal{S}^{d\times d}$ to itself defined via

$$\mathbf{Q}\mapsto f(\mathbf{Q}):=\mathbf{U}^{\mathrm{T}}\mathrm{diag}(f(\gamma_1(\mathbf{Q})),\ldots,f(\gamma_d(\mathbf{Q})))\mathbf{U}.$$

i.e. $f(\mathbf{Q})=\mathbf{U}^{\mathrm{T}}f(\mathbf{\Gamma})\mathbf{U}$, where

$$f(\mathbf{\Gamma})=\begin{bmatrix} f(\gamma_1) & & & \\ & f(\gamma_2) & & \\ & & \ddots & \\ & & & f(\gamma_d)\end{bmatrix}$$

So the eigenvectors are the same, but $\gamma_i\mapsto f(\gamma_i)$.

Example 1.1. Let $\mathbf{Q}$ be a real symmetric $d\times d$ matrix and $f(x)=x^i$, we want to check $f(\mathbf{Q})\overset{?}{=}\underbrace{\mathbf{Q}\cdot\mathbf{Q}\cdots\mathbf{Q}}_{i\text{ times}}$.

By definition,

$$\begin{aligned}\mathbf{Q}^i&=\mathbf{Q}\cdot\mathbf{Q}\cdots\mathbf{Q}\\&=\mathbf{U}^{\mathrm{T}}\mathbf{\Gamma}\underbrace{\mathbf{U}\cdot\mathbf{U}^{\mathrm{T}}}_{=\mathbf{I}_d}\mathbf{\Gamma}\mathbf{U}\cdots\mathbf{U}^{\mathrm{T}}\mathbf{\Gamma}\mathbf{U}\\&=\mathbf{U}^{\mathrm{T}}\underbrace{\mathbf{\Gamma}\cdot\mathbf{\Gamma}\cdots\mathbf{\Gamma}}_{i\text{ times}}\mathbf{U}\\&=\mathbf{U}^{\mathrm{T}}\begin{bmatrix}\gamma_1^i & & & \\ & \gamma_2^i & & \\ & & \ddots & \\ & & & \gamma_d^i\end{bmatrix}\mathbf{U}\\&=f(\mathbf{Q}).\end{aligned}$$

So two definitions coincide.

In particular, the matrix exponential has the power-series expansion,

$$\begin{aligned}
\sum_{k=0}^{\infty} \frac{\mathbf{Q}^k}{k!} &= \mathbf{U}^{\mathrm{T}} \left(\sum_{k=0}^{\infty} \frac{1}{k!} \begin{bmatrix} \gamma_1^k & & & \\ & \gamma_2^k & & \\ & & \ddots & \\ & & & \gamma_d^k \end{bmatrix} \right) \mathbf{U}^{\mathrm{T}} \\
&= \mathbf{U}^{\mathrm{T}} \begin{bmatrix} \sum_{k=0}^{\infty} \frac{\gamma_1^k}{k!} & & & \\ & \sum_{k=0}^{\infty} \frac{\gamma_2^k}{k!} & & \\ & & \ddots & \\ & & & \sum_{k=0}^{\infty} \frac{\gamma_d^k}{k!} \end{bmatrix} \mathbf{U} \\
&= \mathbf{U}^{\mathrm{T}} \begin{bmatrix} e^{\gamma_1} & & & \\ & e^{\gamma_2} & & \\ & & \ddots & \\ & & & e^{\gamma_d} \end{bmatrix} \mathbf{U} \\
&= e^{\mathbf{Q}}.
\end{aligned}$$

The eigenvalues of $e^{\mathbf{Q}}$ are positive, so that it is a positive definite matrix for any choice of $\mathbf{Q}$.

Definition 1.2. A function f on $\mathcal{S}^{d\times d}$ is said to be *matrix monotone* if $f(\mathbf{Q}) \preceq f(\mathbf{R})$ whenever $\mathbf{Q} \preceq \mathbf{R}$.

The question is: Does the exponential function satisfy matrix monotonicity? The answer is NO, showing that matrix monotonicity is more complex than the usual notion of monotonicity. By contrast, the logarithm is that it is a matrix monotone function (HW). Finally, there is a useful inequality.

Lemma 1.3 (trace inequality)**.** *If $f : \mathbb{R} \to \mathbb{R}$ is continuous and non-decreasing function in the usual sense, then for any pair of symmetric matrices such that $\mathbf{Q} \preceq \mathbf{R}$, then*

$$tr(f(\mathbf{Q})) \leq tr(f(\mathbf{R})).$$

2 Tail conditions for matrices

Given a symmetric random matrix $\mathbf{Q} \in \mathcal{S}^{d\times d}$, its polynomial moments, assuming that they exist, are the matrices defined by $\mathbb{E}[\mathbf{Q}^j]$. The variance of $\mathbf{Q}$ is a positive semidefinite matrix given by $\mathrm{var}(\mathbf{Q}) := \mathbb{E}[\mathbf{Q}^2] - (\mathbb{E}[\mathbf{Q}])^2$. The moment generating function of a random matrix $\mathbf{Q}$ is the matrix-valued mapping $\mathbf{\Psi}_{\mathbf{Q}} : \mathbb{R} \to \mathcal{S}^{d\times d}$ given by

$$\mathbf{\Psi}_{\mathbf{Q}}(\lambda) := \mathbb{E}[e^{\lambda \mathbf{Q}}] = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \mathbb{E}[\mathbf{Q}^k].$$

In parallel with our discussion in Chapter 2, various tail conditions are based on imposing bounds on this moment generating function.

Definition 2.1. A zero-mean symmetric random matrix $\mathbf{Q} \in \mathcal{S}^{d\times d}$ is sub-Gaussian with matrix parameter $\mathbf{V} \in \mathcal{S}_+^{d\times d}$ if

$$\mathbf{\Psi}_{\mathbf{Q}}(\lambda) \preceq e^{\frac{\lambda^2 \mathbf{V}}{2}} \qquad \text{for all } \lambda \in \mathbb{R}.$$

This definition is best understood by working through some simple examples.

Example 2.2. Suppose that $\mathbf{Q} = \varepsilon \mathbf{B}$ where $\varepsilon \in \{-1, +1\}$ is a Rademacher variable, and $\mathbf{B} \in \mathcal{S}^{d \times d}$ is a fixed matrix. Since $\mathbb{E}[\varepsilon^{2k+1}] = 0$ and $\mathbb{E}[\varepsilon^{2k}] = 1$ for all $k = 1, 2, \ldots$, and hence

$$\begin{aligned}
\boldsymbol{\Psi}_{\mathbf{Q}}(\lambda) &= \mathbb{E}[e^{\lambda \mathbf{Q}}] \\
&= \mathbb{E}\left[\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \mathbf{Q}^k\right] \\
&= \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \mathbb{E}[\mathbf{Q}^k] \\
&= \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \mathbb{E}[\varepsilon^k \cdot \mathbf{B}^k] \\
&= \sum_{m=0}^{\infty} \frac{\lambda^{2m}}{(2m)!} \mathbf{B}^{2m} \\
&\preceq \sum_{m=0}^{\infty} \frac{\lambda^{2m}}{m! 2^m} \mathbf{B}^{2m} \qquad \left(\frac{1}{(2m)!} \leq \frac{1}{m!\, 2^m}\right) \\
&= \sum_{m=0}^{\infty} \frac{1}{m!} \left(\frac{\lambda^2 \mathbf{B}^2}{2}\right)^m \\
&= e^{\frac{\lambda^2 \mathbf{B}^2}{2}}.
\end{aligned}$$

showing that the sub-Gaussian condition holds with $\mathbf{V} = \mathbf{B}^2 = \mathrm{var}(\mathbf{Q})$.

Lemma 2.3. *If $\mathbf{Q} \in \mathcal{S}^{d \times d}$ with $\|Q\|_2 \leq b$, i.e. $-b\mathbf{I}_d \preceq \mathbf{Q} \preceq b\mathbf{I}_d$, then $e^{\frac{\lambda^2 \mathbf{Q}^2}{2}} \preceq e^{\frac{\lambda^2 b^2 \mathbf{I}_d}{2}}$.*

Proof. Since Q is symmetric and $\|Q\|_2 \leq b$, write $Q = U \,\mathrm{diag}(\mu_1, \ldots, \mu_d) U^\top$, where U is orthogonal and $|\mu_i| \leq b$. Thus,

$$e^{\frac{\lambda^2 b^2}{2}} I_d - e^{\frac{\lambda^2 Q^2}{2}} = U \,\mathrm{diag}\left(e^{\frac{\lambda^2 b^2}{2}} - e^{\frac{\lambda^2 \mu_i^2}{2}}\right)_{i=1}^{d} U^\top \succeq 0,$$

which proves the claim. □

Example 2.4. As an extension of the previous example, consider a random matrix of the form $\mathbf{Q} = \varepsilon \mathbf{B}$, where ε s a Rademacher variable as before, and $\mathbf{B}$ is now a random matrix, independent of ε with its spectral norm bounded as $\|B\|_2 \leq b$. Using previous lemma, we have

$$\boldsymbol{\Psi}_{\mathbf{Q}}(\lambda) = \mathbb{E}_Q[e^{\lambda \mathbf{Q}}] = \mathbb{E}_{\mathbf{B}} \mathbb{E}_\varepsilon[e^{\lambda \varepsilon \mathbf{B}}] \preceq \mathbb{E}_{\mathbf{B}} e^{\frac{\lambda^2 \mathbf{B}^2}{2}} \preceq \mathbb{E}_{\mathbf{B}} e^{\frac{\lambda^2 b^2 \mathbf{I}_d}{2}} = e^{\frac{\lambda^2 b^2 \mathbf{I}_d}{2}} \qquad \text{for all } \lambda \in \mathbb{R},$$

showing that $\mathbf{Q}$ is sub-Gaussian with matrix parameter $\mathbf{V} = b^2 \mathbf{I}_d$.

In parallel with our treatment of scalar random variables in Chapter 2, it is natural to consider various weakenings of the sub-Gaussian requirement.

Definition 2.5. (Sub-exponential random matrices) A zero-mean symmetric random matrix $\mathbf{Q} \in \mathcal{S}^{d \times d}$ is sub-exponential with parameters $(\mathbf{V}, \alpha)$, where matrix parameter $\mathbf{V} \in \mathcal{S}_+^{d \times d}$ and scalar parameter $\alpha \geq 0$, if

$$\boldsymbol{\Psi}_{\mathbf{Q}}(\lambda) \preceq e^{\frac{\lambda^2 \mathbf{V}}{2}} \qquad \text{for all } |\lambda| < \frac{1}{\alpha}.$$

References

[1] M. J. Wainwright, *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*, Cambridge University Press, 2019.